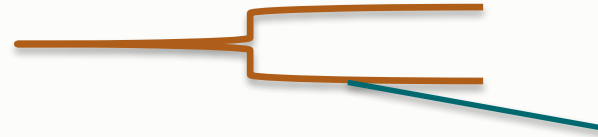


Cluster-robust variance centers uncertainty at the group level

$$\hat{V}_{\text{CR}} = (X^\top X)^{-1} \left(\sum_g X_g^\top \hat{u}_g \hat{u}_g^\top X_g \right) (X^\top X)^{-1}$$



Middle term aggregates score contributions by center.

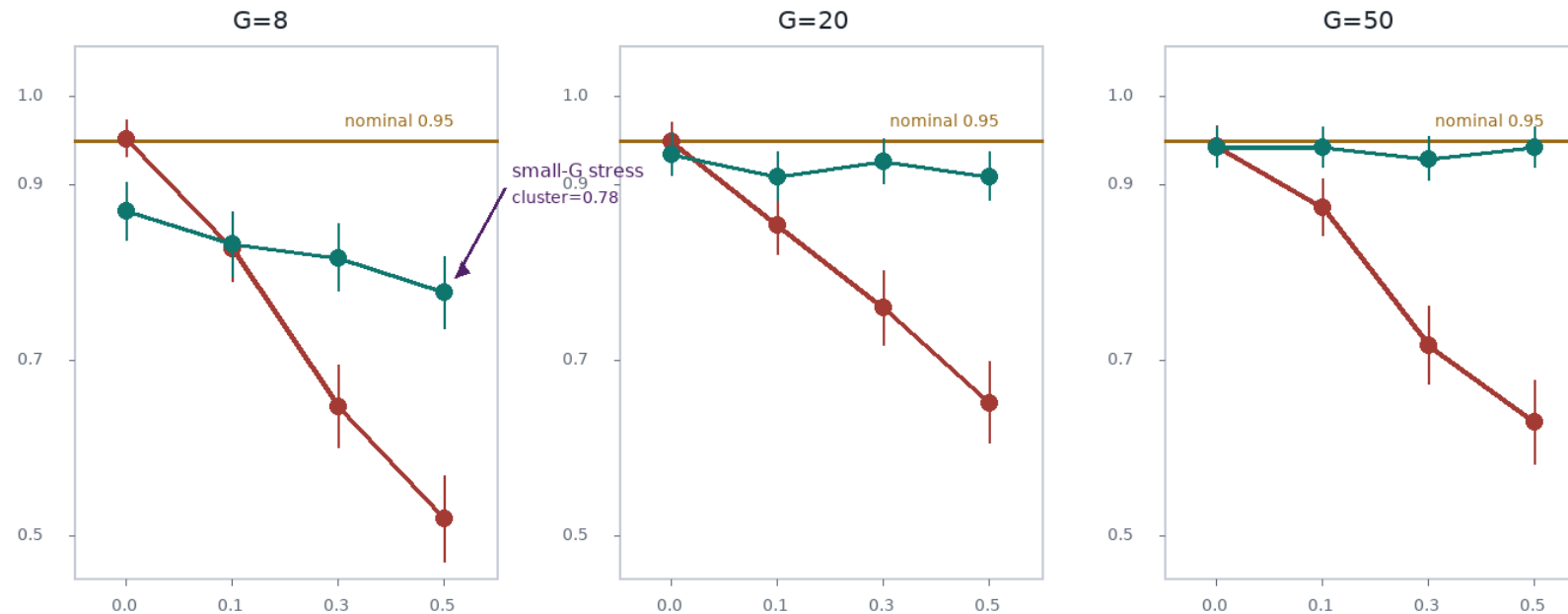
Simulation varies clustering before interval evaluation



DGP knobs feed generated samples and interval procedures.

Cluster-robust intervals recover coverage, but small-G remains stressed

Imbalanced clusters: 95% interval coverage by ICC

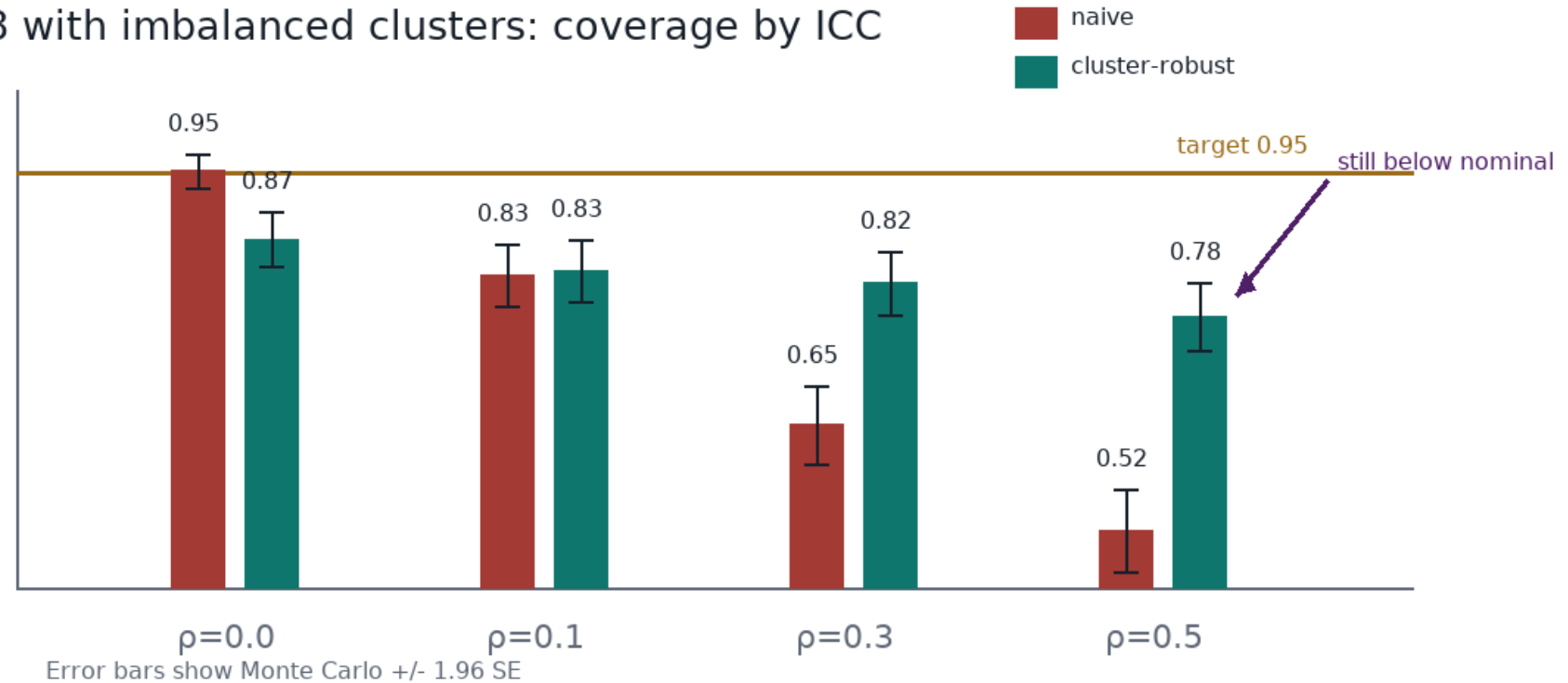


Coverage improves after center-level aggregation, with visible stress at low center counts.

Deterministic simulation; nominal 0.95 reference line shown in the plotted evidence.

Small center counts remain anti-conservative

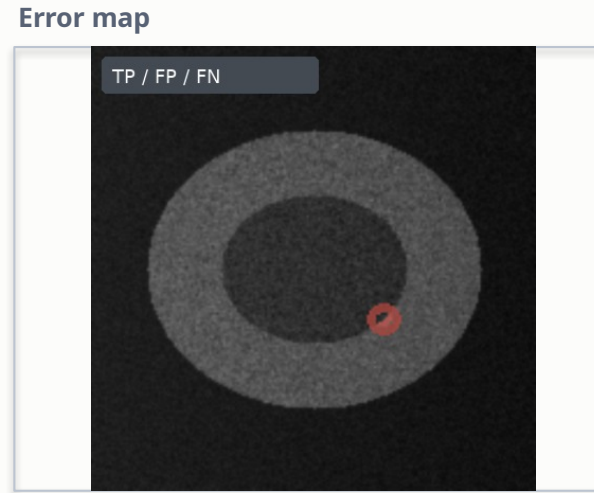
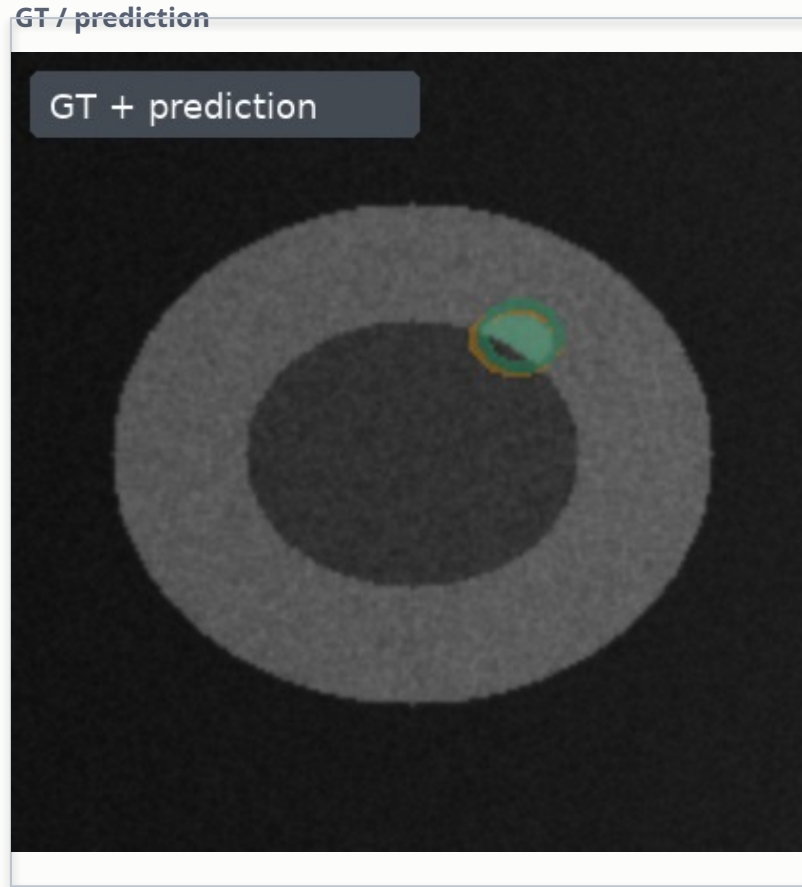
G=8 with imbalanced clusters: coverage by ICC



The failure region is the small-G corner; the next experiment should target finite-sample correction.

Negative result motivates CR2 or wild cluster bootstrap as the next discriminating experiment.

Overlay errors localize the small-lesion failure mode



The miss is spatially small but changes lesion-level recall.

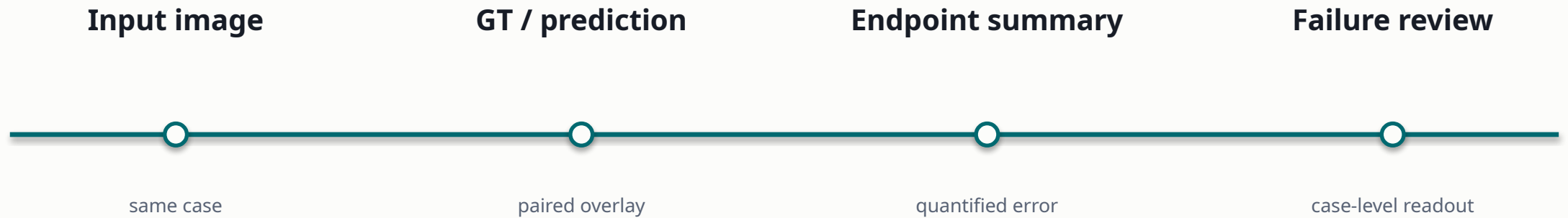
■ overlap

■ false positive

■ false negative

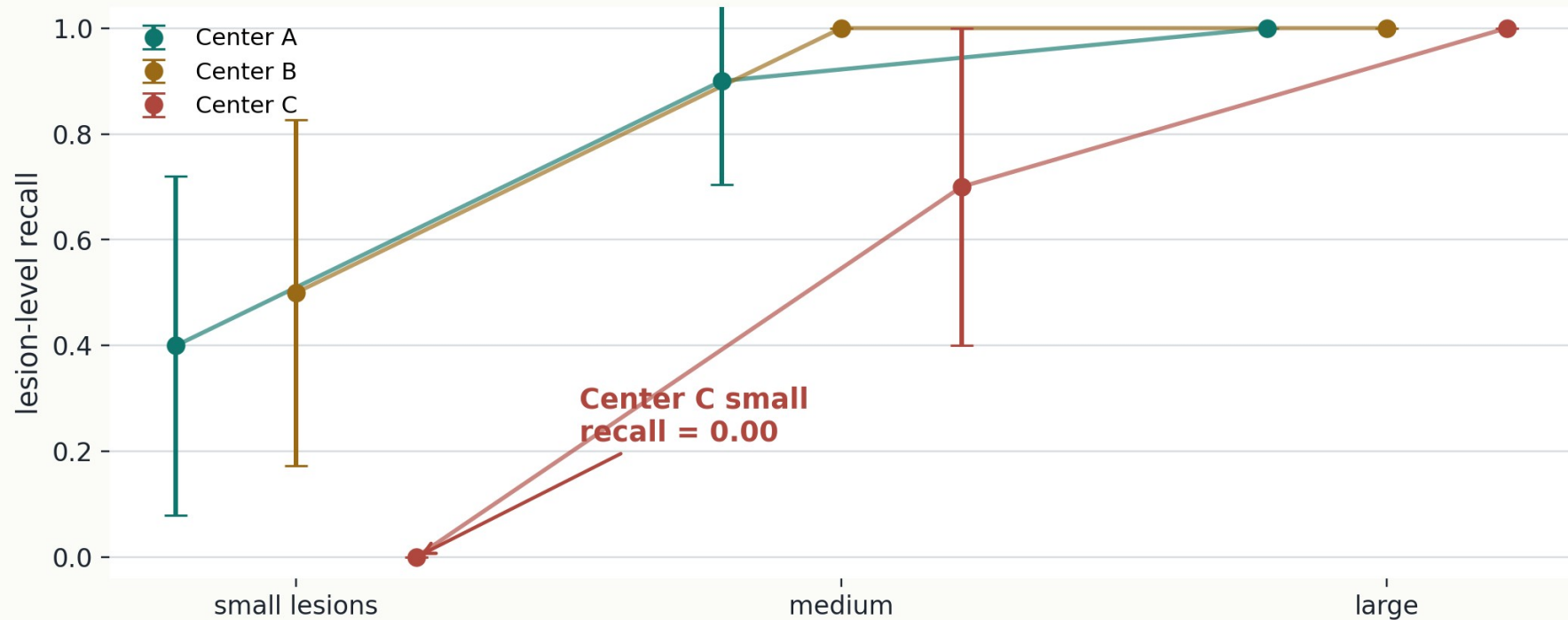
Synthetic public-safe image fixture; image treatment is integrated, clinical realism remains out of scope.

The evaluation path keeps image evidence and endpoints paired



Input, mask, prediction, and endpoint summaries are generated from the same synthetic case.

Lesion-size recall exposes the endpoint-level weakness



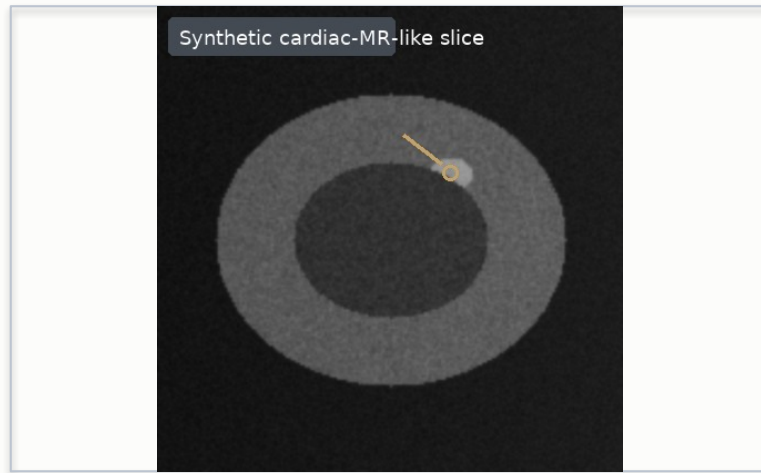
Completed evidence: deterministic synthetic phantoms only; next validation is planned, not completed.

Recall drops most sharply for the smallest lesions in this deterministic fixture.

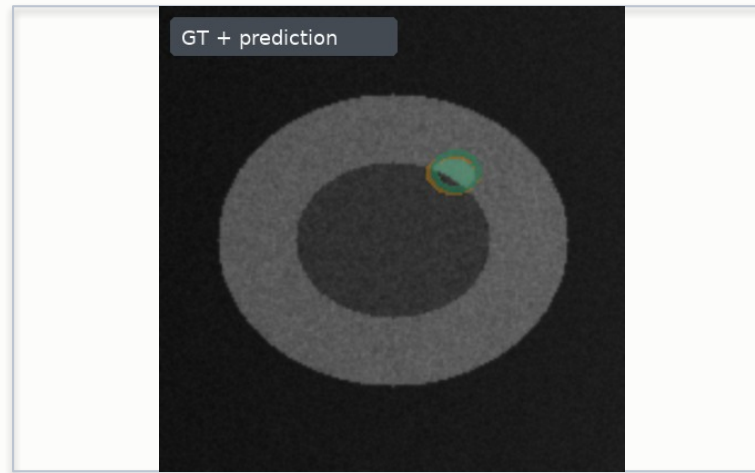
Deterministic fixture result; not clinical validation.

Same-case error maps show where the endpoint is lost

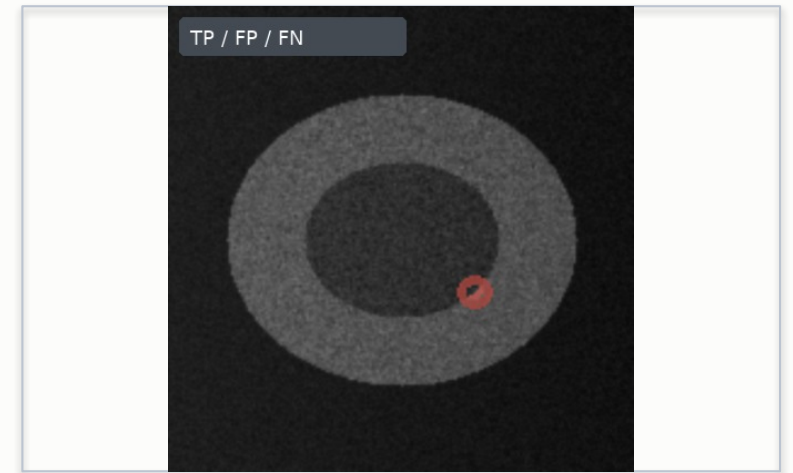
Input



GT / prediction



Error map



The miss is spatially small but changes lesion-level recall.

■ overlap ■ false positive ■ false negative

Synthetic public-safe image fixture; image treatment is integrated, clinical realism remains out of scope.